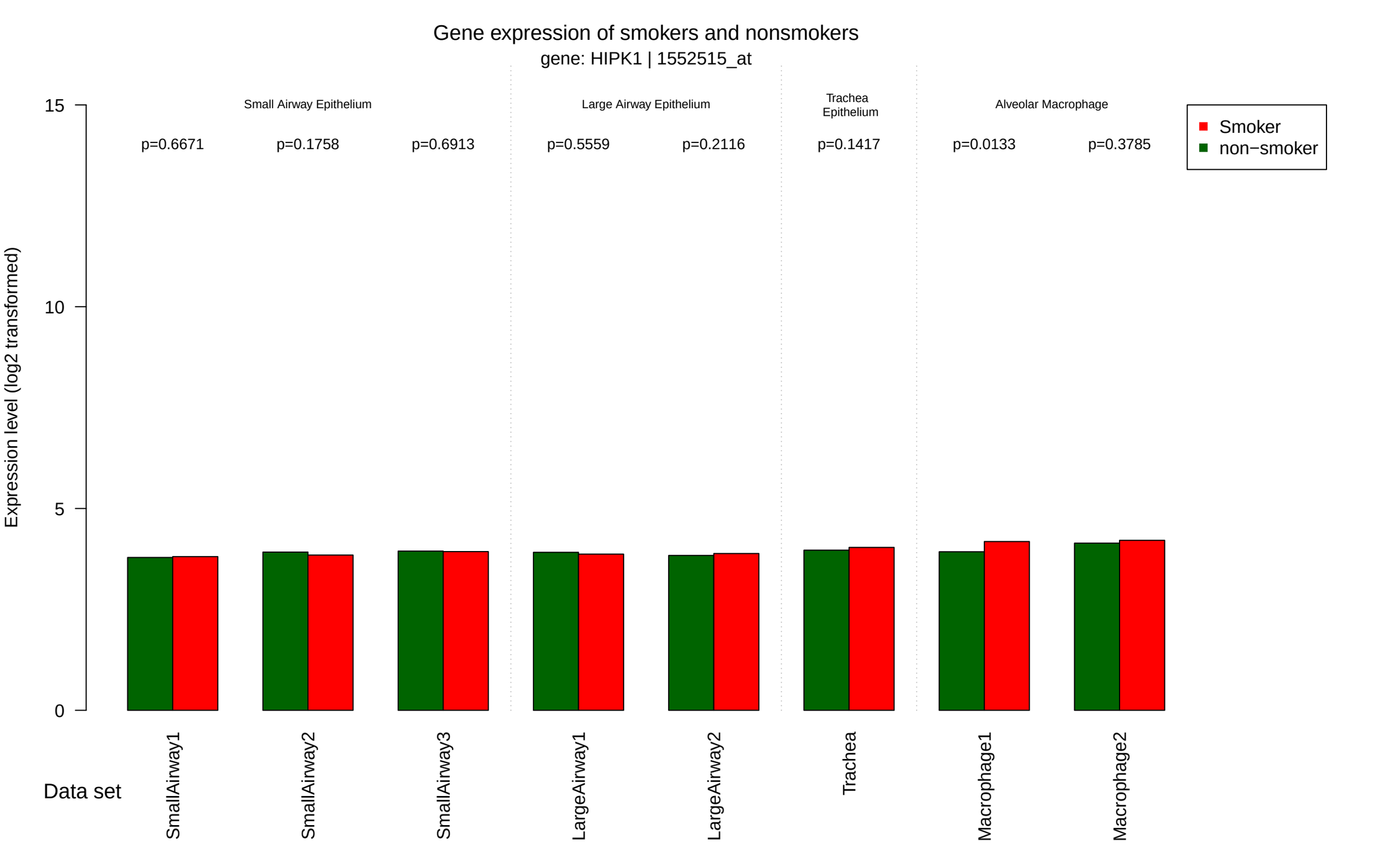
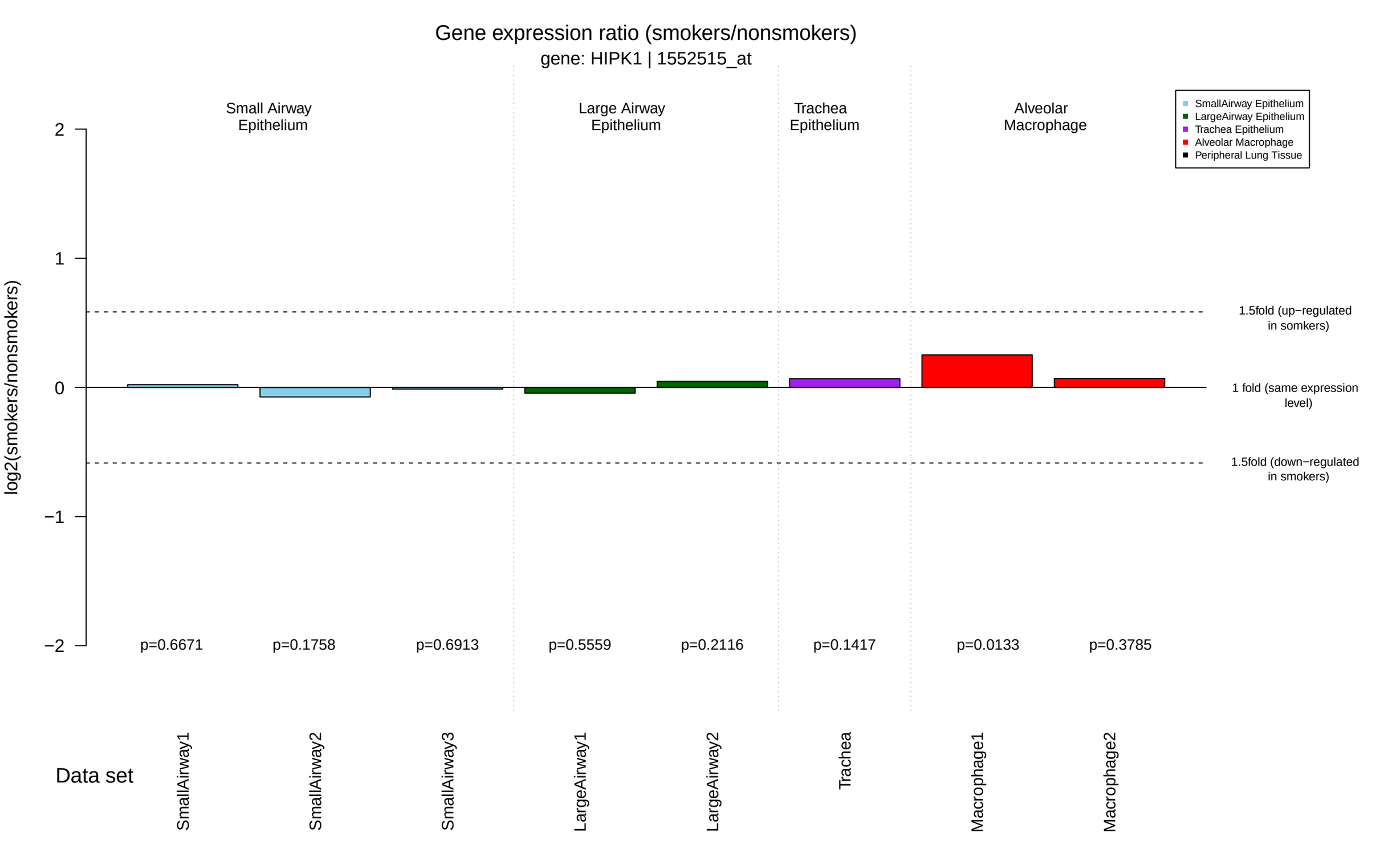
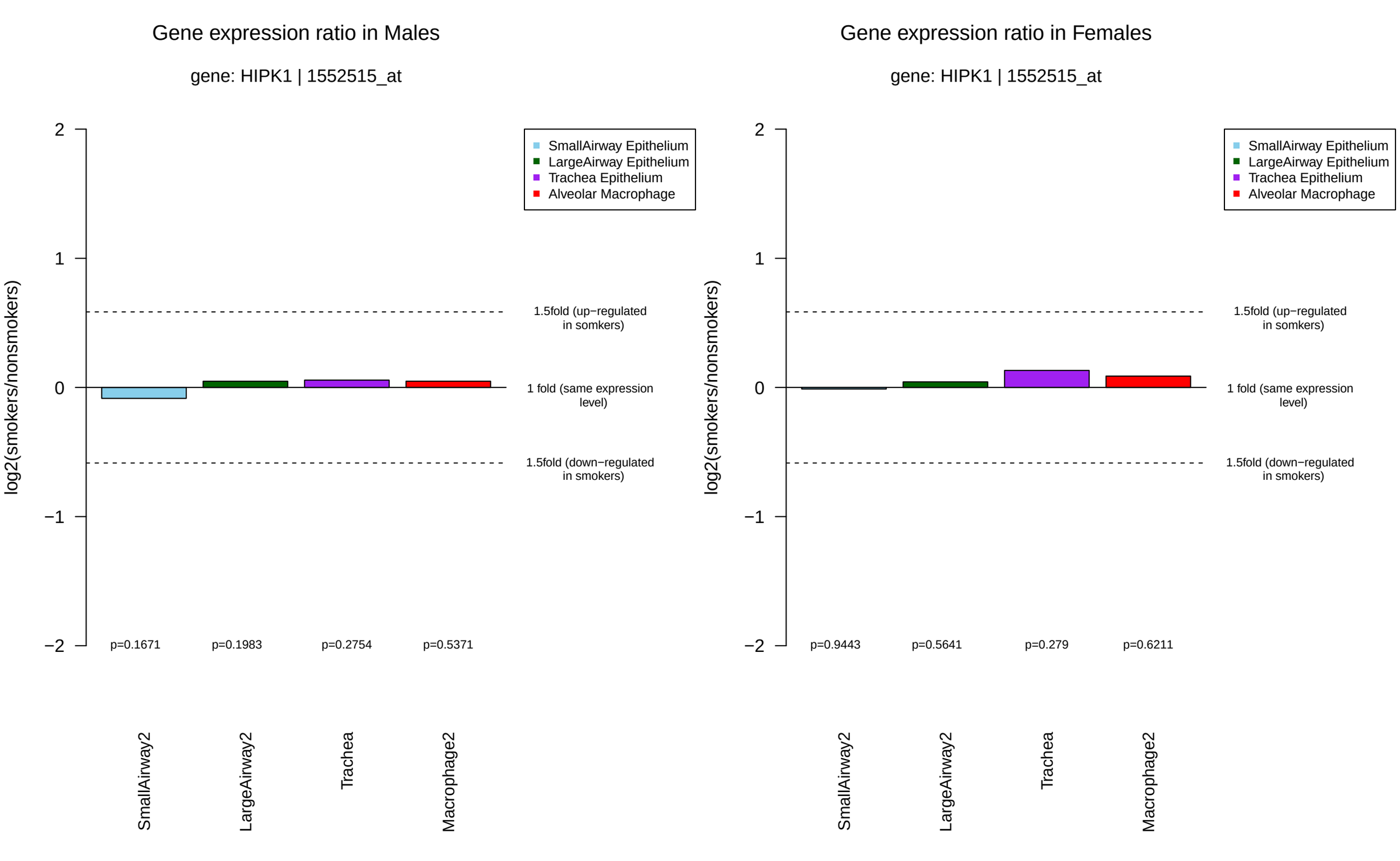
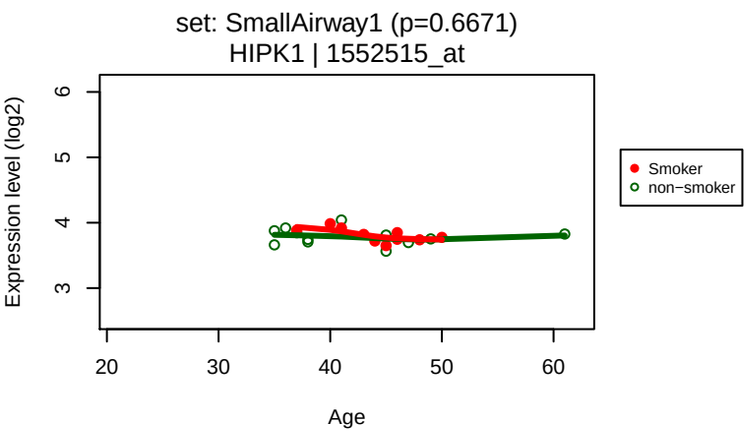
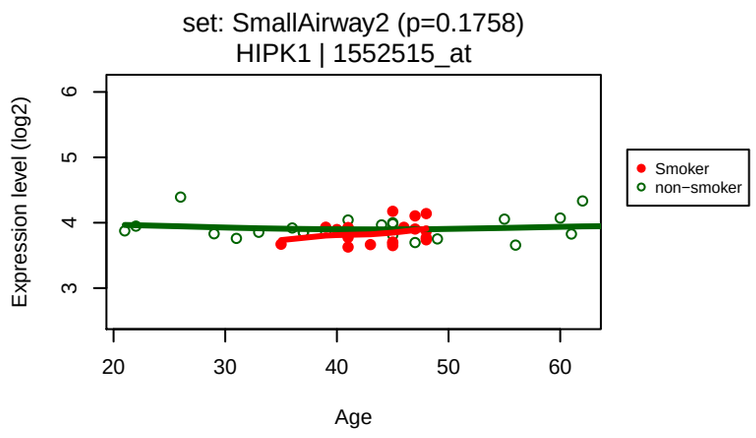
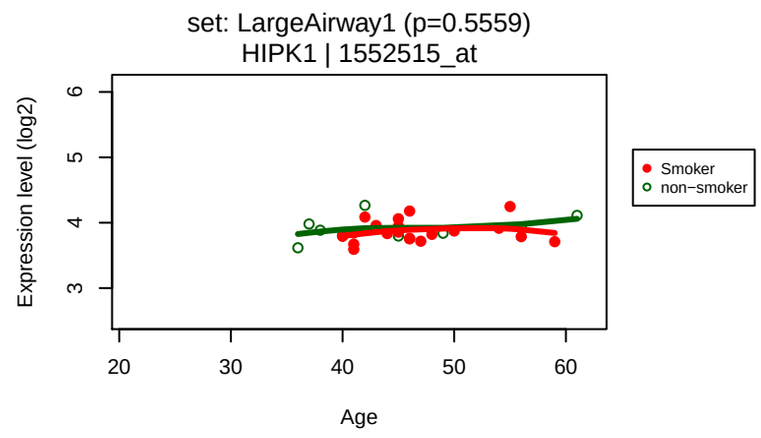

Gene expression of smokers and nonsmokers
gene: HIPK1 | 1552515_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6671
p=0.1758
p=0.6913
p=0.5559
p=0.2116
p=0.1417
p=0.0133
p=0.3785
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

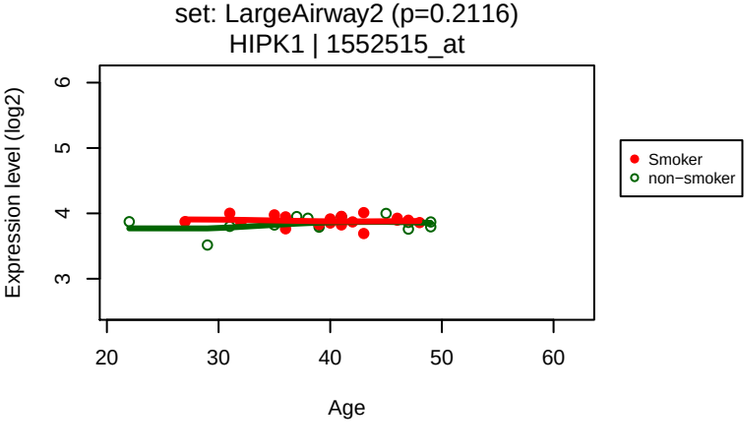

Gene expression ratio (smokers/nonsmokers)
gene: HIPK1 | 1552515_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6671
p=0.1758
p=0.6913
p=0.5559
p=0.2116
p=0.1417
p=0.0133
p=0.3785
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

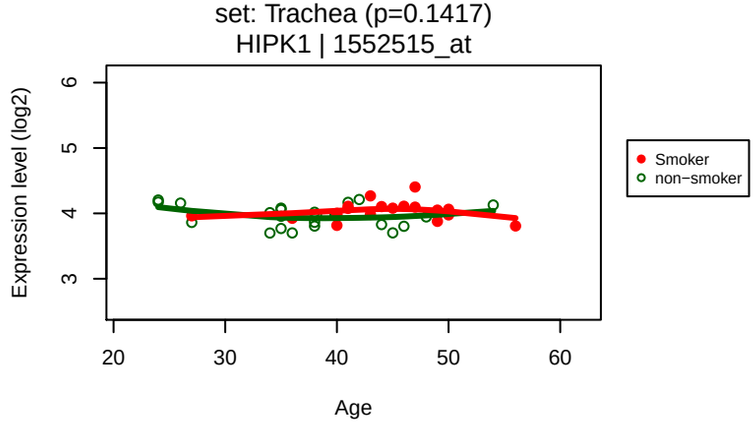

Gene expression ratio in Males
gene: HIPK1 | 1552515_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1671
p=0.1983
p=0.2754
p=0.5371
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK1 | 1552515_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9443
p=0.5641
p=0.279
p=0.6211
SmallAirway2
LargeAirway2
Trachea
Macrophage2

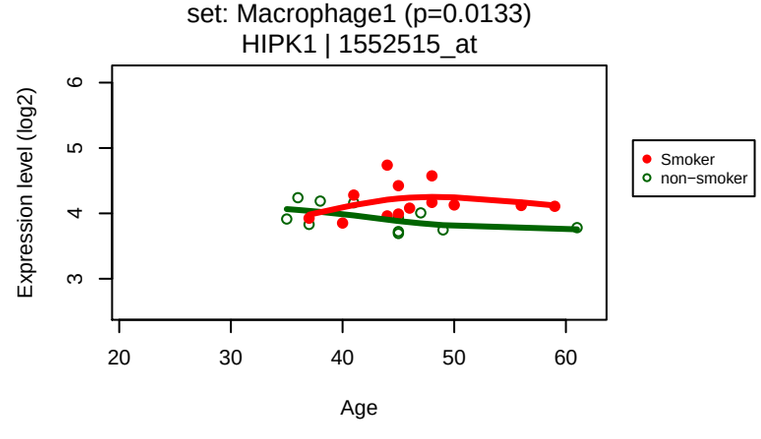

set: SmallAirway1 (p=0.6671)
HIPK1 | 1552515_at
Expression level (log2)
Age
Smoker
non−smoker

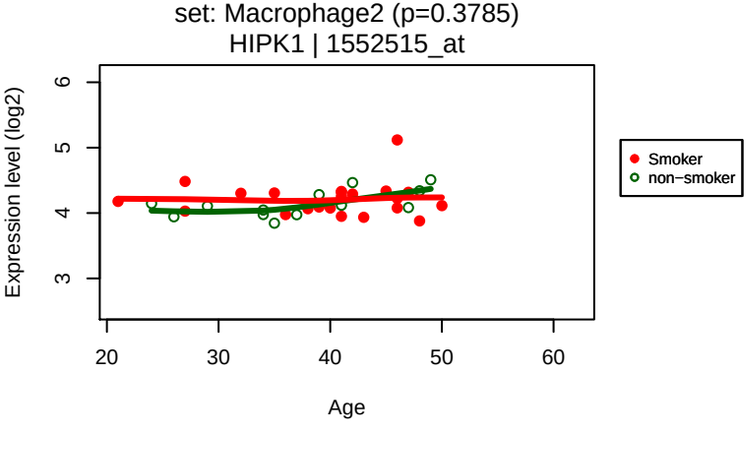

set: SmallAirway2 (p=0.1758)
HIPK1 | 1552515_at
Expression level (log2)
Age
Smoker
non−smoker

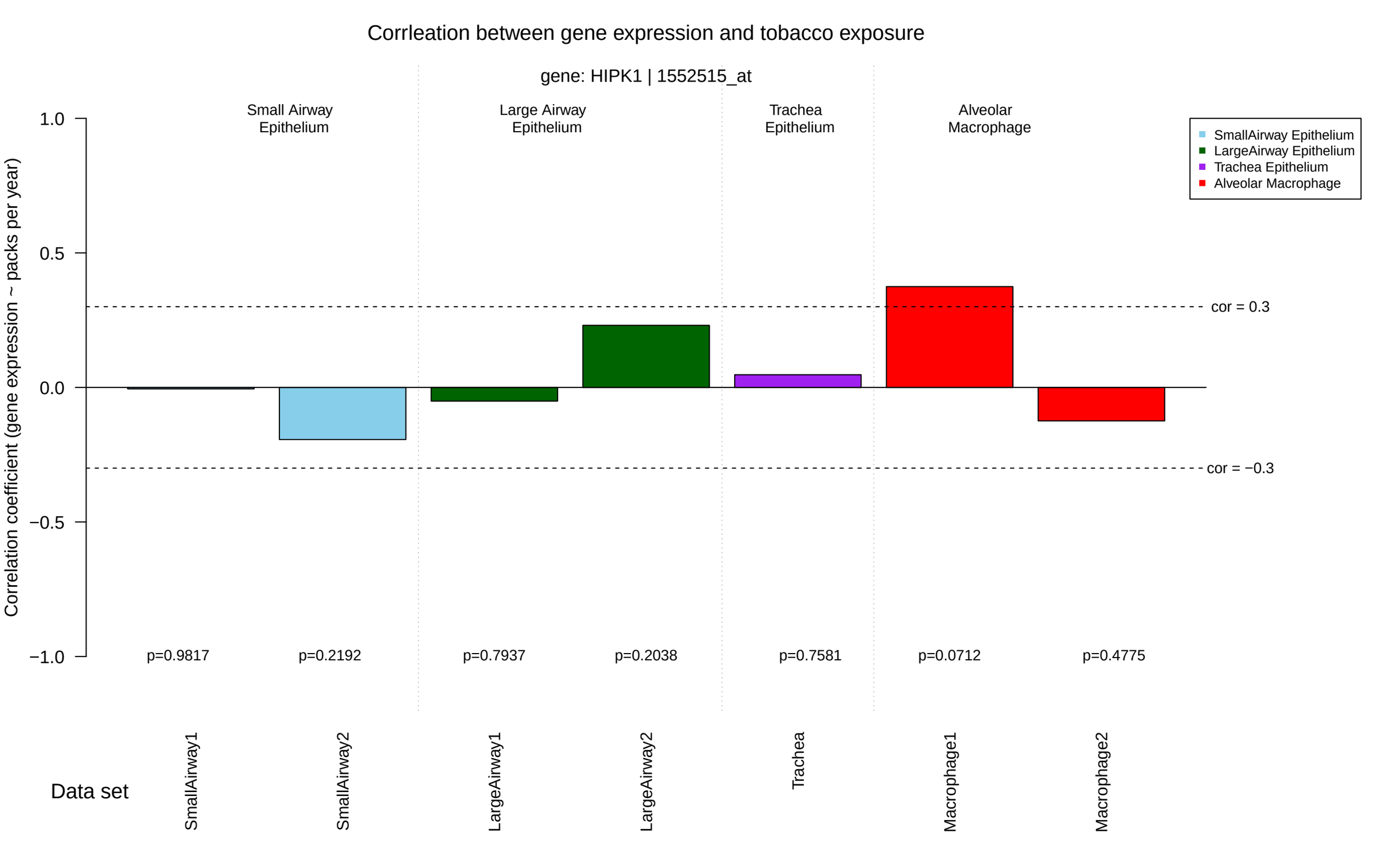

set: LargeAirway1 (p=0.5559)
HIPK1 | 1552515_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2116)
HIPK1 | 1552515_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1417)
HIPK1 | 1552515_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0133)
HIPK1 | 1552515_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3785)
HIPK1 | 1552515_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK1 | 1552515_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9817
p=0.2192
p=0.7937
p=0.2038
p=0.7581
p=0.0712
p=0.4775
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2